Exoskeleton Bring-Up and Validation Runbook

Oct 24, 2026 · @Tharaka

Read this first

Work through the phases in order. Each phase ends with a gate, and you do not move on until the gate passes, because every later step assumes the earlier ones are true.

The runbook takes the assembled prototype from first power-up (24 Oct 2026) to a filled results tracker and a frozen report (27 Nov 2026). It uses your validation programme (Tests 0 to 11) as the backbone and adds the practical steps around them that no test table lists: inspection, wiring checks, safe limits, logging, backups.

Each step is written the same way:

- **Why now.** The reason this step sits here and not later.
- **How.** What to do, in order.
- **Pass.** The result that lets you move on (taken from your test criteria where one exists).
- **If not.** What to check first, in the order that finds the fault fastest.
- **Time.** A working estimate, not a promise.

Assumptions behind the timing (change the dates if any is wrong):

- You can work about 5 to 6 focused hours a day, Monday to Saturday; Sunday is rest or catch-up.
- You run control, electronics and testing; the mechanical member helps with the shell, the rig and the transmission; the ML member supplies the replayed EMG.
- The control loop runs on the PC at a 500 Hz target, the two STS3215 servos sit on one bus, and the joint angle comes from the AS5600 encoder.
- Where your protocol already fixes a method, follow it. Where it does not, the method written here is a proposal; adjust it and record what you changed.

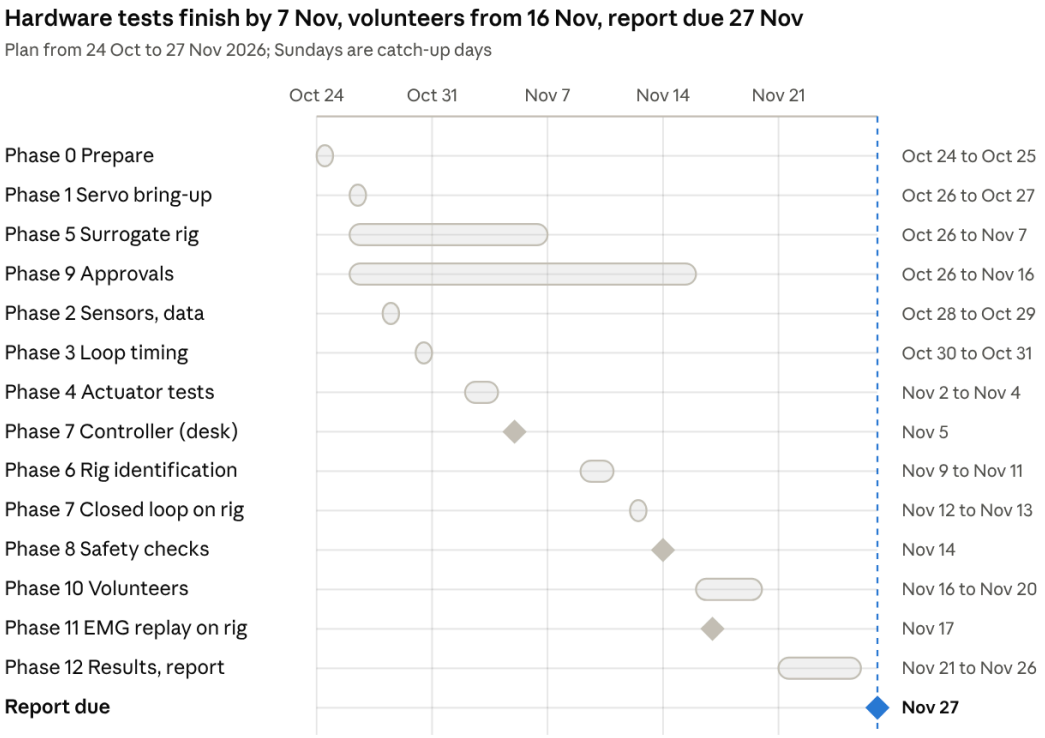
Rules that apply to every step:

1. **Change one thing at a time.** If two things change between runs and the result moves, you cannot tell which one did it.
2. **Log every run.** One file per run, named `YYYYMMDD_HHMM_testname_subject_Kd.csv`, with a header holding the code version, the settings, finger posture and forearm rotation, and who ran it.
3. **Nothing reaches a person before it has passed on the surrogate rig.** This is D-16, and it is also the cheapest way to find your own mistakes.
4. **Keep a hand on the power cut whenever the motors are live with anything attached.**
5. **Write the result down the moment you have it, including failures.** A failed test with its recovery action is a result; an unrecorded run is lost work.
6. **Photograph the setup before each test** (strap positions, wiring, masses). You will need it when a number looks odd a week later.
7. **Back up at the end of every day** to a second place (cloud folder or second drive).

8. **If the same step fails twice for the same reason, stop.** Go to the troubleshooting map at the end instead of retrying.

Timeline

Hardware is validated on the bench by 7 Nov, the rig tests and safety checks finish on 14 Nov, volunteers come from 16 Nov, and the last week is for results and the report, due Fri 27 Nov.



plan from 24 Oct to 27 Nov 2026 · 13 phases, report due 27 Nov

Gates in the calendar: G3 timing passes (Fri 30 Oct); G4 actuator tests and model updated (Wed 4 Nov); G7 closed loop on the rig (Fri 13 Nov); G8 safety table passes (Sat 14 Nov); G9 readiness checklist and approval (Mon 16 Nov). The plan assumes 5 to 6 focused hours a day, Monday to Saturday, and has almost no slack; the section at the end lists what to cut first.

Phase 0: Prepare before any power (Sat 24 to Sun 25 Oct)

Gate G0: the device passes inspection, the electronics pass the power-off checks, the workspace and software are set up, and the long-lead requests are sent.

Step 0.1 Inspect the assembled device with the power off

Why now. Most first-day damage comes from a mechanical or wiring mistake that a careful look would have caught.

How.

1. Compare the device with the drawings and bill of materials: every fastener present, the servo horn secure, no loose screws in the cable path.
2. Check the cable loop: correct routing, no rubbing on an edge, equal pre-tension on both sides, knots or crimps secure.
3. Check the AS5600 magnet: centred over the chip, the gap within the datasheet range, magnet firmly fixed.
4. With the servos unpowered, move the joint by hand through the full range (±45°). Feel for binding, slack that suddenly goes taut, or a scraping sound.
5. Mark the mechanical neutral with tape and note where the hard stops are.

Pass. The joint moves smoothly through the full range and stops where you expect.

If not. Binding: disconnect the cable loop from the servo and move the joint alone. If it is now free, the transmission is binding; if it still binds, the joint bearing or axis is misaligned. Sudden slack or tightening: the cable is routed over an edge or one pulley is eccentric.

Time. 1 hour.

Step 0.2 Check the electronics with the power off, then with a current limit

Why now. A short or reversed supply destroys a servo or the PCB in a fraction of a second; a multimeter finds it for free.

How.

1. With nothing powered, measure resistance from each supply rail (12 V, 5 V, 3.3 V) to ground on the PCB. None should read close to zero.
2. Confirm connector polarity against the schematic for the 12 V input and the servo bus cable.
3. Set the bench supply to 12 V with the current limit at 0.3 A. Power the PCB only, with the servos and sensors unplugged. Measure each rail: within ±5% of nominal.
4. Plug in the sensors one at a time, checking the rail and the supply current after each.
5. Confirm the servo supply ground and the USB adapter ground are common.

Pass. All rails within ±5%, supply current steady and small, nothing warm.

If not. A rail is wrong or the current sits at the limit: power off at once, remove the last part you plugged in, and find the fault with the continuity test. A component that is warm to the touch within a minute is the suspect.

Time. 1 to 2 hours.

Step 0.3 Set up the workspace

Why now. The first time the motors move, the device must not be able to walk off the bench or injure anyone.

How. Clamp the device base firmly to a stable bench. Put an inline switch (the E-stop) in the 12 V line within arm's reach. Use a bench supply with an adjustable current limit. Keep cables clear of moving parts. Have a second person in the room for any run with the device powered and attached to something.

Pass. You can cut motor power with one hand without looking.

Time. 1 hour.

Step 0.4 Set up software, folders and version control

Why now. Every later result must be traceable to the exact code and settings that produced it.

How.

1. Install the USB-serial driver for the servo adapter and the sensor board; note the COM port names.
2. Create a Python environment with pinned versions and the serial and servo-bus libraries.
3. Create the folders: `control/src`, `control/config`, `data/raw` (never edited after saving), `data/processed`, `docs`.
4. Put every constant (transmission radii, limits, M_d, K_d levels, sample rates) in one config file. No magic numbers inside the code.
5. Start a git repository; commit before every test session and note the commit in the log header.
6. Create the results tracker (one row per test: date, code version, result, pass or fail, file name).

Pass. A fresh clone of the repository runs the empty program with no manual fixes.

Time. 2 to 3 hours.

Step 0.5 Gather tools and spares

Multimeter; bench supply with current limit; oscilloscope or USB logic analyser if you can borrow one (it helps for timing); calibrated masses and a scale; calipers and a tape measure; a dial indicator or a fixed angle reference; a goniometer (Test 11 needs it later); spare Dyneema cable, spare fasteners and thread-locker; labels and a camera.

Time. 1 hour, plus any borrowing.

Step 0.6 Start the long-lead items today

Why now. These take days of other people's time and gate the human tests in Phase 10.

- **Approval.** Ask your supervisor what approval and consent are required to test on healthy volunteers (including team members), and lodge the request on Monday 26 Oct.
- **Surrogate rig.** Ask the mechanical member to start the rig described in Phase 5 and order the spring and masses now.
- **Replay EMG.** Ask the ML member for the file format and one sample recording (needed in Phase 11).

Time. 1 hour of messages.

Phase 1: First power-up and servo bring-up (Mon 26 to Tue 27 Oct)

Gate G1: both servos are found on the bus, have unique labelled IDs, move safely inside their limits in both axes with the transmission attached, and you know what happens when power is cut.

Step 1.1 Find the servos on the bus with the transmission disconnected

Why now. The first commands you send are the ones most likely to be wrong (wrong ID, wrong direction, wrong units). If the cable loop is detached, a wrong command moves an empty horn instead of a joint.

How.

1. Detach the cable loop from the servo, or free the servo horn from the joint.
2. Set the bench supply to 12 V with the current limit at 1 A. Power one servo at a time on the bus adapter.
3. Scan the bus with the library (or the manufacturer's debug tool). Read the model number, firmware version, supply voltage, temperature and present position.
4. If both servos still share the factory ID, connect one at a time and give them different IDs (for example one per axis). Write the ID on a label stuck to each servo.
5. Record the baud rate and IDs in the config file.

Pass. Both servos answer a ping, report a voltage near 12 V and a temperature near room temperature, and have unique labelled IDs.

If not. No answer: check the baud rate first (scan all rates with the debug tool), then the half-duplex adapter wiring (data line, direction control), the servo supply, the common ground and the ID. Answers only sometimes: reseat the connectors, shorten the cable, and watch whether the supply voltage sags when the servo moves.

Time. 1.5 hours.

Step 1.2 Listen before you speak: read-only logging

Why now. It shows the noise floor and any communication errors before any movement can hide them.

How. With torque disabled, log position, voltage and temperature at the rate you plan to use for 5 minutes. Count read errors and timeouts.

Pass. No read errors at the planned rate; position noise within a step or two (0.088° per step).

If not. Errors that appear at higher rates only: lower the rate and note the limit. Errors at any rate: wiring, termination, or electrical noise near the servo cable (route it away from the motor supply lines).

Time. 30 minutes.

Step 1.3 First movements, unloaded, with limits set first

Why now. Limits must exist before the first command, not after the first surprise.

How.

1. Write the angle limits (±45° from neutral) into the servo's own registers if it has them, and into the software config. Use both.
2. Set low speed and low acceleration. If the servo exposes a torque limit or protection setting, set a low value for now.
3. Enable torque and command ±5°, then ±20°, then slowly to each limit and back.

4. Watch direction, smoothness and the supply current. Record the current during motion and at rest.

Pass. The servo reaches each commanded position within a few steps, in the expected direction, without buzzing or hunting, and stops at the limits.

If not. Wrong direction: flip the sign in the config, never in the middle of the code. Buzzing at hold: a mechanical load or backlash, or supply sag; leave the servo's internal gains at their defaults for now. A position offset: set the neutral in the config and re-check. It overshoots the limit: the limit registers are not applied; read them back to confirm.

Time. 1 hour.

Step 1.4 Find out what happens when things go wrong

Why now. Every safety argument later depends on knowing what the hardware does when power or communication is lost.

How. With the servo unloaded, test each case and write down the result:

- Press the E-stop while the servo holds a position: does the horn go free or stay locked?
- Stop the PC program mid-motion: does the servo hold its last target?
- Unplug the bus cable mid-motion: same question.
- Release torque by command while holding a small load: does the load fall freely?

Pass. You can state for each case what the joint does, and it is acceptable for a person (the E-stop must remove torque).

If not. If the servo keeps driving to a stale target after the PC dies, then the E-stop is your only fail-safe in that case. Keep targets rate-limited so a stale target is never far from the present position, and add the watchdog in Phase 8.

Time. 30 minutes.

Step 1.5 Reattach the transmission

Why now. Only now is it safe to connect a joint to the servo, because the servo is known and limited.

How. Reattach the cable loop with the servo at neutral and the joint at its marked neutral. Set the pre-tension equally on both sides. Move through the full range slowly and watch the cable, the clamps and the supply current.

Pass. The joint follows the servo through ±45° with no slack, no slipping and a smooth current.

If not. Slipping or skipping: increase pre-tension or fix the clamp. Rising current near one end: the cable is binding or the mechanical stop is reached before the software limit; reduce the limit or fix the stop. Noise or jerks: check the pulleys for flats or burrs.

Time. 1 hour.

Step 1.6 Repeat for the second axis and record baselines

Repeat Steps 1.1 to 1.5 for the other axis. Then run both axes for 10 minutes and note the servo temperatures, the supply current and the load register readings while free-moving. Save these as the baseline for later comparison.

Time. 3 hours for the second axis and the baselines.

Phase 2: Sensors and data path (Wed 28 to Thu 29 Oct)

Gate G2: every angle channel reads correctly, the units and signs are fixed in the config, the two sample streams share one time base, and a 10-minute log saves with no gaps.

The controller, the torque estimator and every test read these channels. If a sensor is wrong by a sign or a scale factor, every later result is wrong in a way that looks plausible. This is the cheapest phase in which to find that.

Step 2.1 Bring up the joint encoder (AS5600) on its own

Why now. It is the ground truth for the joint angle theta_j in every test.

How.

1. Read the raw 12-bit value and the magnet status flags (magnet detected, too weak, too strong).
2. Turn the joint slowly by hand through ±45° and plot the reading. It must rise or fall smoothly with no jumps.
3. Handle the 0 to 4095 wrap-around in code (unwrap), even if the joint never crosses it, because a different mount might.
4. Define the zero at the marked neutral and the positive direction (document it in the config).
5. Hold the joint still for 60 seconds and compute the standard deviation of the reading.

Pass. Smooth, monotonic over the full range; magnet flags report "detected", not weak or strong; noise of only a few counts (one count is 0.088°).

If not. Weak or strong magnet flag: change the gap between magnet and chip. A curve that bends or wobbles over one revolution: the magnet is off-centre. Noisy reading: check the decoupling capacitor on the sensor supply, route the sensor cable away from motor wires, shorten it. Occasional jumps: I2C errors (check pull-up resistors and cable length) or unhandled wrap-around.

Time. 2 hours.

Step 2.2 Cross-check the servo encoder against the joint encoder

Why now. The torque estimate on RUD is the difference between the two angles, so their ratio must be right before Test 0 can mean anything.

How.

1. With the transmission attached, move the servo slowly from −45° to +45° and back, logging servo angle theta_m and joint angle theta_j together.
2. Plot theta_j against theta_m. Fit a straight line through the middle of the range.

3. Compare the fitted slope with the ratio of the radii in your drawings (r_m / r_j).
4. Look at the gap between the forward and backward passes: that is backlash plus cable twist.

Pass. A straight line whose slope agrees with the geometric ratio to within a few percent. The remaining difference is the cable twist and backlash, which Test 0 will measure properly.

If not. Slope wrong by a fixed factor: a wrong radius in the config, or the cable slipping on the pulley. Large gap between passes: slack in the cable loop, so re-tension. A line that bends: an eccentric pulley or a cable that wraps differently across the range.

Time. 1.5 hours.

Step 2.3 Fix the time base and the sample rates

Why now. The servo encoder gives 200 Hz and the joint encoder 500 Hz. If they do not share a clock, the difference between them (the torque estimate) contains timing error as well as twist.

How.

1. Timestamp every sample on one clock (the PC's monotonic clock at the moment of arrival, or a device timestamp mapped to it).
2. Log 5 minutes and histogram the intervals between samples of each channel.
3. Decide how the two rates are merged: hold the last servo value and store its age with every joint sample. Do not interpolate silently.

Pass. Mean interval within 1% of nominal for each channel, no dropped samples in 5 minutes, and the age of the servo value is always below one servo period plus jitter.

If not. Occasional long intervals: USB buffering or the operating system scheduling. Close other programs, raise the process priority, use a fixed-size binary protocol, and keep printing out of the loop. Drifting intervals: two clocks being compared; use one.

Time. 2 hours.

Step 2.4 Fix the units and signs once

Why now. Sign and unit mistakes are the most common reason a controller works in simulation and fights the hardware.

How. Move the servo by one count and confirm 0.088°. Convert counts to radians in one function only. Write down: positive theta_j means (for example) flexion or radial deviation; positive torque means the same direction. Check that positive servo motion gives positive joint motion. Put a short comment block in the config file with these conventions.

Pass. A person who has not seen the code can predict the direction of every sign from the comment block.

Time. 30 minutes.

Step 2.5 Look at the servo load register (flexion/extension torque proxy)

Why now. Flexion/extension has no twist-based torque estimate; it uses the load register, which is uncalibrated and a declared limitation (D-24). You should know how well it behaves before relying on it.

How. With the joint held by the servo, push on the joint by hand in each direction and log the load register. Then hang a known mass on a known lever arm and record the register for several loads, increasing and decreasing.

Pass. The register changes monotonically with load, has the same sign for the same direction, and the up and down passes roughly agree. Record the result as it is; no calibration is claimed.

If not. Saturation or heavy noise: filter it, record the usable range, and state the limitation in the report instead of hiding it.

Time. 1.5 hours.

Step 2.6 Test the whole logging pipeline

Why now. The first time you discover a logging bug must not be in the middle of a test you cannot repeat.

How. Run a 10-minute log of all channels at the planned rates while moving the joint by hand. Check the row count against the expected count, confirm the header is written, open the file in a plotting tool and look at every channel.

Pass. No gaps, header complete, every channel plots sensibly. Back it up.

Time. 1 hour.

Phase 3: PC loop timing, Test 7 (Fri 30 to Sat 31 Oct)

Gate G3: Test 7 passes (worst-case tick under 2 ms over 60 s), or the recovery is applied and the test passes again; the transport delay T_d is measured and written into the config.

Why this comes before any controller. The architecture (D-22) puts the control loop on the PC. Whether a PC can hold 500 Hz through a USB bus to two servos is an open question until you measure it, and every stability margin you compute later uses this timing. If it fails, you want to know before you tune anything.

Step 3.1 Write an empty loop that does everything except control

How.

1. Build a fixed-rate loop that uses absolute deadlines: each tick's start time is the previous start plus one period, not "sleep for one period". This stops errors from accumulating.
2. Inside it: read the joint encoder, read the servo position (and load), write a servo goal equal to the present position (so nothing moves), and timestamp each stage.
3. Log, for every tick, the start time, the duration of each stage and the total.
4. Keep printing and plotting out of the loop; log to memory and write to disk after the run.

Pass. The loop runs for 60 seconds at 500 Hz and produces a timing log.

Time. 3 hours.

Step 3.2 Run Test 7

How. Run 60 seconds at 500 Hz with your normal logging on and nothing else heavy running. Compute the maximum tick duration, the 99th and 99.9th percentiles, the number of ticks over 2 ms, and the jitter of the start times. Repeat three times, then once for 10 minutes. Then repeat with a realistic background load (browser, plotting program open) to see how fragile it is.

Pass. Worst-case tick under 2 ms over 60 s, in all three repeats.

If not, in this order:

1. **Find the slow call.** Look at the stage timings. If the servo read or write dominates, use the library's grouped (sync) read and write for both servos instead of one request each.
2. **USB-serial latency.** Some adapters wait up to 16 ms before delivering data by default. Set the adapter's latency timer to its minimum (1 ms) in the driver settings.
3. **Operating system timing.** On Windows the default timer tick is coarse; raise the timer resolution, set the process to high priority, turn off power saving, and busy-wait for the last millisecond before each deadline.
4. **Garbage collection pauses** (Python): preallocate arrays, avoid creating objects inside the loop, or pause the collector during the run.
5. **Still failing:** apply the recovery in your test table: read the servos at 250 Hz and keep the joint-encoder loop at 500 Hz. Record the change and re-run Test 7.
6. **Fails even then:** move the fast part (reading and filtering) to the microcontroller on your PCB and keep only the supervisory loop on the PC. This is a design change, so tell your supervisor and record it in the decision log.

Time. 2 hours if it passes first time; up to a day if you have to go down the list.

Sub-test: time skew between the two angle sensors. The servo encoder updates at 200 Hz (every 5 ms) and the joint encoder at 500 Hz (every 2 ms), and the torque estimate is the difference between them. A sample that is up to 5 ms old shows up as a false torque while the wrist moves: false torque = angular velocity × age × k_theta. With the device unloaded and no person in it, move the joint by hand at several speeds and plot the difference between the two angles against the speed. The slope is the effective time skew in milliseconds. Note that the speeds you try, and k_theta from Test 0, decide how large the false torque is; I did not use measured values when I sketched this. If the skew is more than about 1 ms, fix it before closed-loop work: run the admittance loop at 200 Hz, or timestamp every sample and interpolate the joint angle to the instant of the latest servo reading. Re-run the plot to confirm the slope falls to near zero.

Step 3.3 Measure the transport delay T_d

Why now. The loop equation contains exp(−s T_d), and T_d is currently an assumed 5 ms.

How.

1. Time the bus round trip alone: send a read request, record when the reply arrives (software timestamps, or a logic analyser on the bus lines if you have one).

2. Command a small step in goal position and find when the joint angle first moves beyond the noise. Repeat 20 times and take the median.
3. Remember the joint encoder samples every 2 ms, so the delay can only be resolved to about that; use many repeats and report the spread.

Pass. T_d between 2 and 8 ms (your criterion).

If not. Above 8 ms: shorten the path (grouped reads, lower USB latency, smaller buffers). Below 2 ms: check that the step really starts the timer when the command is sent and not when the loop tick begins. Either way, the loop analysis later uses the measured value, not 5 ms.

Time. 2 hours.

Step 3.4 Build the timing guard now

Count overruns in the loop. If a tick exceeds the limit, log it; if several happen in a row, command the safe state (hold, then release torque). Testing this properly is part of Phase 8, but putting the counter in now means every later run is monitored.

Time. 1 hour.

Step 3.5 Record the decision

Write the loop rate, the servo read rate and the measured T_d into the config and the results tracker (Test 7 row), with the date and code version. If you changed the architecture or the rate, add an entry to the decision log.

Time. 30 minutes.

Phase 4: Actuator and transmission tests (Mon 2 to Wed 4 Nov)

Gate G4: Tests 6, 8, 0 and 5 are recorded (passed, or recovered and passed), the measured values are in the config, and the stability rules have been re-checked with them.

The order matters. The shell rigidity test goes first because a flexing shell corrupts every angle and torque you will measure afterwards. The servo step response comes next because the loop analysis needs it. Test 0 needs a trustworthy angle reference, so it follows.

Step 4.1 Test 6: shell rigidity (go/no-go)

Why now. If the forearm shell twists under load, the encoder and the torque estimate read the shell flexing as joint motion. Nothing downstream can be trusted until this passes.

How.

1. Clamp the device base to the bench exactly as it will be worn (straps and clamps included).
2. Apply a known torque through a lever arm: a mass m on an arm of length r gives torque m g r. For example 1.02 kg at 0.10 m gives about 1.0 N·m.
3. Measure the twist of the shell relative to a fixed external reference with a dial indicator at a known radius (twist in radians = displacement divided by radius). Do not use the joint encoder for this, because it sits on the shell.

4. Load and unload three times; record the reading at 0, 0.5 and 1 N·m.
5. Keep fingers clear of the hanging mass and do not exceed about 1 N·m.

Pass. Under 1.6° at 1 N·m (that is K_s of at least 36 N·m/rad); target under 0.5°.

If not. Find where the flex is before stiffening anything: bridge each joint in turn with a rigid block and repeat. Most flex comes from the strap and clamp interfaces and loose fasteners, not the shell wall. Then stiffen: tighten, add ribs, thicker walls, a stiffer material or a better clamp. Repeat the test. Do not continue with Phase 4 until it passes.

Time. 3 hours.

Step 4.2 Test 8: servo step response

Why now. The loop analysis uses the servo's bandwidth f_s and damping zeta_s, currently assumed (8 Hz, 0.8).

How.

1. Transmission attached, hand plate fitted, no person. Set the servo's speed and acceleration registers to their maximum for this test (otherwise the servo's own ramp, not its dynamics, is what you measure). Record the register values.
2. Command steps of 2°, 5° and 10° from a few starting angles. Log the joint angle at 500 Hz for at least 1 second after each step. Repeat each 5 times.
3. Fit a second-order response with the measured delay T_d: from the overshoot OS, damping zeta = −ln(OS) / sqrt(pi^2 + ln(OS)^2); from the peak time t_p, the damped frequency is pi / t_p, and the natural frequency f_s follows from that and zeta. If there is no overshoot, fit the rise time instead.
4. Compare the three step sizes. If the result changes a lot with size, the servo is not behaving linearly (deadband, friction, speed limit).

Pass. f_s between 5 and 15 Hz and zeta_s between 0.5 and 1.2.

If not. The motor model in the stability analysis is wrong, so recompute every margin with the measured values (that is your recovery). Then look for the cause. f_s too low: speed or acceleration limit still active, supply voltage sagging during the step, friction or a slack cable. zeta_s too low (ringing): compliant or creeping cable, loose coupling, or the servo's own gains; re-tension first. Results that differ between step sizes: deadband or friction; report it.

Time. 3 hours.

Step 4.3 Test 0: transmission stiffness k_theta on RUD

Why now. The torque estimate on RUD is k_theta times the difference between servo-side and joint angles. k_theta is that gain.

How.

1. Hold the servo at a fixed position. Fix the joint end so it can still rotate only through the transmission.

2. Apply known torques with weights on a lever, from 0 up to about 1 N·m in 6 to 8 steps, then back down. At each step wait a few seconds for creep to settle, then average about one second of data of both angles.
3. Compute the difference Delta = theta_m (r_m / r_j) − theta_j at every step.
4. Plot torque against Delta. The slope is k_theta. Compute R^2, and the hysteresis as the largest gap between the up and down curves divided by the full range.
5. Repeat three times, then leave the cable loaded for 10 minutes and repeat once to see creep (Dyneema creeps).

Resolution check. At k_theta of 100 N·m/rad, 1 N·m gives a difference of 0.01 rad, which is about 0.57° or 6 to 7 counts. That is measurable but coarse, so use the largest safe torque, average many samples and state the uncertainty.

Pass. R^2 of at least 0.98, hysteresis under 15%, and k_theta between 30 and 250 N·m/rad.

If not (your recoveries). Too stiff (above 250): add a series spring, because the difference would be too small to measure. Too soft (below 30): increase pre-tension. Curved fit: a slipping clamp or a cable running over an edge. Wide up and down loop: re-tension and wait for creep to settle.

Time. 4 hours.

Step 4.4 Test 5: coupling between the two axes

Why now. The design treats flexion/extension and radial/ulnar deviation as independent loops. This checks that assumption on the hardware.

How. Use the c_12 definition in your test protocol (a ratio of cross-axis response to driven-axis motion). Drive one axis through a slow sweep while the other axis is held by its servo, log the other axis's sensor, and compute the ratio. Do it both ways. Flexion/extension uses the load-register proxy.

Pass. c_12 below 0.10.

If not. Model the coupling (Jacobian), or move one axis at a time; mode-switched operation already does this. Also check for mechanical causes first: an axis that is not perpendicular, a cable that drags on the other axis.

Time. 2 hours.

Step 4.5 Put the measured numbers into the model and re-check the rules (desk work)

Why now. Until now the loop equation used assumed constants. This is the step where the design meets the hardware.

How.

1. Put the measured f_s, zeta_s, T_d, k_theta and K_s in the config.
2. Recompute the loop gain L(s) = [Z_env / Z_d] · G_servo · exp(−s T_d) at the corners of the expected ranges: J_h 0.001 to 0.006, K_h 0.43 to 1.89, B_h 0.005 to 0.08, together with the extremes of the measured servo values. Check the margins at every K_d level.
3. Re-check the two rules for every level and write the result as a small table (rule met, not met).

Something to catch now. With the design M_d of 0.010 kg·m², the rule $M_d \geq 3 J_h$ holds only for J_h up to about 0.0033, but the expected range runs to 0.006. For a subject with a heavy hand, M_d would have to rise to about 0.018. Because $K_d = M_d (2\pi f_n)^2$, raising M_d at fixed f_n also raises K_d (about 2.8 N·m/rad at 2 Hz), so you will either accept the higher K_d or lower f_n. Decide the policy now, as part of the per-subject rule check, rather than discovering it with a person wearing the device.

Pass. A table of K_d levels against the ends of the J_h and K_h ranges, each marked as allowed or not allowed.

Time. 3 hours.

Phase 5: The surrogate rig (build from Mon 26 Oct; characterise Fri 6 to Sat 7 Nov)

Gate G5: the rig has a written datasheet giving J, B and K with how each was obtained and its uncertainty, it is mounted on the device with the axes aligned, and it swings freely.

Why a rig at all. A wrist is an unknown. A rig whose J, B and K you know is the one place where you know the right answer. It lets you check the hardware, the identification code and the controller before a person is involved (D-16). It also acts as a negative control: it has no spasticity, so Tests 4 and 2b should show none.

Step 5.1 Specify and build the rig (mechanical member, in parallel with Phases 1 to 4)

Why now. It is a physical item with parts to order, so it has to start in week 1 or it will hold up week 3.

What to ask for.

- **A stiff arm and hand-plate analogue** that mounts on the wrist axis of the device through the same interface the hand plate uses, and rotates about the same axis.
- **Interchangeable springs** so K spans the expected wrist range (0.43 to 1.89 N·m/rad). A linear spring acting at a lever arm r gives $K = k r^2$ for small angles.
- **Mounting points for two equal masses** at a known radius, for Test 1b. For example two 100 g masses at 80 mm give $2 m r^2 = 1.28 \times 10^{-3}$ kg·m², the added inertia in your test table.
- **A viscous damper** (for example a paddle in a viscous fluid) adjustable to give B in the expected range (0.005 to 0.08 N·m·s/rad). Dry friction is not viscous damping; avoid it.
- **Hard stops and guards** so a fault cannot hurt anyone or break the rig.

Configuration	Spring	Added masses	Used for
A	soft (about 0.5 N·m/rad)	none	low-K end; K_d levels 0.4 and 0.8 rule check
B	middle (about 1 N·m/rad)	none	nominal case
C	stiff (about 1.9 N·m/rad)	none	high-K end
D	as B	two masses at known radius	Test 1b added-inertia check

Time. The mechanical member's time; you review the drawing (1 hour).

Step 5.2 Characterise the rig (do not trust the label)

How.

1. **J.** Calculate it from the mass and geometry of the arm, and check it with a pendulum swing (period T = 2 pi sqrt(J / (m g d)) for a known mass m and centre distance d). The added inertia comes from 2 m r^2 with weighed masses and measured radius.
2. **K.** Hang known weights on a lever arm and measure the angle for 5 to 6 loads, up and down. The slope of torque against angle is K. Do it for each spring.
3. **B.** Let the rig swing freely and fit the decay of the oscillation (the logarithmic decrement gives the damping ratio, and B follows from J and K). B is the least certain of the three; say so in the datasheet.
4. **Uncertainty.** For each of J, B and K write how it was obtained and how uncertain it is. Do not claim more traceability than you measured.

Pass. A one-page datasheet per configuration with J, B, K, method and uncertainty.

If not. If K depends on the angle (the spring is not linear over the range), restrict the use range or choose another spring. If B changes between runs, the damper fluid is warming or the paddle is not fully immersed; fix before using it as a reference.

Time. 1 day.

Step 5.3 Mount the rig and check it

How. Fit the rig to the device wrist axis. Check that the two axes are collinear (a misalignment makes the rig push sideways and corrupts the torque estimate). With the servo torque disabled, swing the rig by hand and confirm it moves freely, returns to rest, and the joint encoder reads sensibly. Confirm the stops engage before any damage point.

Pass. Free swing, collinear axes, clean encoder reading, stops engage first.

Time. 2 hours.

Step 5.4 Test 9, stage A: remount repeatability of the hardware on a rigid phantom (Fri 6 to Sat 7 Nov)

Why. Test 9 asks how small an angle change you can believe after the device is taken off and put back on. You cannot see 1° by eye, and the encoder is not the limit: one count is 0.088°, so 1° is about 11 counts, and quantization alone contributes only about 0.025° (0.088 / sqrt(12)). The limit is the physical reference: strap slip, mount play, and how well the neutral is found. A person adds skin and soft-tissue movement on top. On a rigid phantom those are removed, so what remains is the hardware itself. If the hardware alone cannot reach 1°, a person cannot either, so do this before any volunteer.

How.

1. Use the rig plate or a rigid forearm and hand model with a hard stop that fixes neutral. Check the stop with a digital inclinometer. Choose one whose datasheet accuracy is clearly better than 0.3° (check the datasheet; 0.3° is a target I suggest, not a figure from a source).

2. Fit the device and run the neutral-offset (zeroing) step exactly as written in the procedure. Hold at the stop and log the joint encoder for 5 seconds; take the mean.
3. Remove the straps and the mount completely and refit them. Photograph the strap positions each time. Repeat at least 10 times (20 is better).
4. Compute the standard deviation of the mean angles across mounts. This is the standard error of measurement (SEM). The 95% minimum detectable change is MDC = 1.96 × sqrt(2) × SEM. For MDC of 1° you need an SEM of about 0.36° or less.
5. Report the uncertainty of the number itself. The standard deviation from N mounts is uncertain by about 1 / sqrt(2(N - 1)): about 24% at N = 10 and 16% at N = 20. Give a confidence interval (a chi-square interval on the standard deviation), not just one value.

Pass. The MDC and its confidence interval are reported. The target is 1° or less on the phantom. This is a design goal that I proposed; it is not taken from a standard or a paper. A result whose interval straddles 1° is reported as "not distinguishable from 1°", not as a pass.

If not. Find the largest source first. Strap slip: mark the strap with a pen and compare before and after. Zeroing: is neutral set by a stop or by eye? Mount play: wiggle the mount by hand and watch the angle. Encoder: check the magnet gap and the AS5600 reading at rest. Fix the largest one and repeat. Report the real number anyway.

Time. 90 minutes for 10 remounts and the analysis.

Phase 6: Identification tests on the surrogate (Mon 9 to Wed 11 Nov)

Gate G6: Tests 1, 1b, 2, 2b, 3, 4 and 4b are run on the rig, the analysis code returns the rig's known J, B and K within their stated uncertainty, and every result is in the tracker.

Why before the controller. These tests are how you will measure a real wrist. Running them on a rig whose answer you know checks the device, the sensors, the torque estimate and your analysis code all at once. A wrong answer here is a bug you can find; a wrong answer on a person is only a doubt.

Step 6.0 Prepare the analysis first

Why now. Analysis code that has never met known data cannot be trusted with unknown data.

How.

1. Write each analysis script and test it on synthetic data: generate a damped oscillator or a noisy ramp with chosen J, B, K, run the script, and confirm it returns those values.
2. Add the noise level you measured in Phase 2 to the synthetic data and see how much the answer moves.
3. Run a dry run of each test without analysing it, to check the log is complete.
4. Warm the servos for 5 minutes before each session, and re-check the neutral zero.
5. Write the rig configuration and settings into every log header.

Pass. Every script recovers the synthetic parameters within a few percent.

Time. 4 hours (can be done during Phase 4 and 5 waiting time).

Step 6.1 Test 1: free oscillation

How.

1. Rig configuration B. Displace to about 10 to 15° with the servo holding, then release the servo torque so the rig swings freely.
2. Record at least 5 seconds at 500 Hz. Repeat 5 times.
3. Fit a damped sinusoid to the angle, skipping the release transient. This gives the natural frequency and damping ratio. With K from Test 2: J = K / omega_n² and B = 2 zeta sqrt(K J).
4. Compute the spread over the 5 trials.

Check before you start. Confirm the rig still swings freely with the servo torque off. If the servo and gears load the swing, the test measures the device plus the rig; compare with the rig alone from Step 5.2 to see how large the device's contribution is, and report it.

Pass. Fit R^2 of at least 0.95; spread over 5 trials under 20%; J within a few percent of the rig's known J.

If not (your recovery, then the order to check). Low R^2 or spread over 20%: repeat, do not average. Surrogate mismatch means the fault is in the method or the code, so no human data until fixed. Check in this order: sign and units; time base; the start of the fit window (the release transient); dry friction (it makes the decay linear, not exponential, so look at the shape of the envelope); the servo loading the swing; and last the rig datasheet itself.

Time. 3 hours.

Step 6.2 Test 1b: added-inertia oscillation

How. Configuration D (two equal masses at a known radius). Repeat Test 1. With the natural frequencies w (without) and w' (with the added inertia delta-J = 2 m r^2): J = delta-J · w'^2 / (w^2 − w'^2) and K = J w^2. Weigh the masses and measure the radius again; the answer depends on them directly.

Pass. delta-J = 1.28×10^{-3} kg·m^2 (±0.26%), and J and K agree with Tests 1 and 2 within 15%.

If not. A disagreement is information. Check the mass radius, the inertia of the cross arm, strap shift and, on a person, relaxation. Note that $w^2 - w'^2$ is a difference of close numbers if the added inertia is small compared with J, so it amplifies noise; use a delta-J comparable to J.

Time. 2 hours.

Step 6.3 Tests 2 and 2b: slow ramp, stiffness and end-feel

How.

1. Command the servo to move the rig slowly (5°/s or less) across about ±15° and back. At that speed the damping torque is tiny compared with the spring torque (for B of 0.03 N·m·s/rad at 5°/s it is about 0.003 N·m), so the torque is almost purely elastic.
2. Take torque from the estimator (k_theta times the angle difference). Plot torque against joint angle. The slope is K and the zero-torque angle is theta_0. The area between the up and down curves is the hysteresis.
3. Repeat 3 times for each spring configuration.

4. Test 2b: from the same curves compute K at the end of the range divided by K in the middle (the end-feel index).

Pass. K between 0.2 and 3.0 N·m/rad and consistent with the rig's static K; the loop area reported. Test 2b is reported only, with no pass or fail.

If not. K far off the rig's value: the error in k_theta carries straight into K, so recheck Test 0 first. A wide loop: friction in the rig, creep in the cable, or a ramp faster than 5°/s.

Time. 3 hours.

Step 6.4 Test 3: pseudo-random perturbation

How.

1. Command small steps in position (about 1 to 2°) with a pseudo-random binary sequence whose bit time is around 25 to 50 ms, so most of the power lies below 10 Hz. Record at least 60 seconds, several repeats.
2. Use the measured angle (not the commanded one), because the servo's bandwidth limits what actually moves.
3. Estimate the impedance from the cross-spectrum of torque and angle, with the coherence. Fit J, B, K.

Pass. Coherence of at least 0.8 over 1 to 10 Hz; J, B, K within 25% of Tests 1 and 2.

If not. Low coherence: amplitude too small (noise dominates), the window too short, or nonlinearity (friction, backlash) in the rig. Disagreement over 25%: treat it as a finding and check posture, mount and transmission coupling first.

Time. 3 hours.

Step 6.5 Tests 4 and 4b: velocity ramps and repeated stretch (null tests on the rig)

Why on a rig. The rig has no spasticity and no tissue softening. So both tests should show nothing. That makes them a negative control for the measurement chain.

How.

1. **Test 4.** Run ramps at several speeds (random order, two passes each, at least 60 seconds of rest between passes, as D-17 requires). Subtract the passive torque predicted from J, B, K. The residual should be close to zero at every speed.
2. **Test 4b.** Ten consecutive passes at one fixed speed; record the peak resisting torque per pass.
3. **Check the speed first.** The servo may not hold a constant speed for long enough at the top of the range (3 rad/s covers 45° in about 0.26 s). Use the measured speed, not the commanded one, and state the highest speed you actually achieved.

Pass. Residual and trend are consistent with zero on the rig. The patient-protocol limit (monotonic residual, threshold 0.2 to 3 rad/s; decay with R^2 of at least 0.9) applies to people.

If not. A residual that grows with speed: the passive model is wrongly subtracted (check B) or there is a velocity-dependent error in the estimator. A trend across the ten passes: thermal drift of the servo, cable creep, or settling of the rig; this points at the measurement chain.

Time. 3 hours.

What the rig should show

Test	Expected on the rig	If different, suspect first
1	J, B, K within the rig's stated uncertainty	sign or units, fit window, servo loading the swing, the datasheet
1b	Added inertia recovered; J and K within 15% of Tests 1 and 2	masses, radius, cross-arm inertia
2	K within the rig and k_theta uncertainty; small loop	k_theta, creep, friction, ramp too fast
2b	Index close to 1 (a linear spring)	spring nonlinearity, hitting a stop
3	Coherence 0.8 or more over 1 to 10 Hz	amplitude too small, window too short, backlash
4	Residual close to zero at all speeds	wrong B subtracted, servo cannot hold the speed
4b	No trend over 10 passes	drift, cable creep, settling

Phase 7: The controller, offline first then on the rig (desk Thu 5 Nov; hardware Thu 12 to Fri 13 Nov)

Gate G7: the closed loop on the rig matches the prediction at every K_d level the stability rules allow, and a 30-minute run shows no timing overruns, drift or overheating.

Why this order. A control law has three kinds of bug: maths, sign and timing. The desk catches maths, a dry run with the output disabled catches signs, and only then does anything move. Moving a rig with a known answer catches the rest.

Step 7.1 Write the controller and test it without hardware (Thu 5 Nov)

How.

1. **Admittance filter.** Integrate M_d x'' + B_d x' + K_d x = tau_ext with x = theta_ref − theta_0 using a semi-implicit Euler step at the loop period (2 ms). At the design point (natural frequency 2 Hz) the step is far inside the stable region.
2. **One tick, in this order:** read the sensors, compute the torque estimate tau_hat (RUD: k_theta times the angle difference; FE: the load register with its declared scaling), form tau_ext = tau_hat + tau_intent (tau_intent bounded to ±0.8 N·m, zero for now), step the filter, compute theta_ref,

then the safety stack: 8 to 10 Hz low-pass, rate limit, ±45° clamp, anti-windup, then write the servo goal.

3. **Anti-windup.** When the clamp or rate limit is active, set the filter state back to the clamped value so it does not keep integrating a request that cannot be met.
4. **Unit tests.** A constant tau_ext gives a steady state x = tau_ext / K_d. A step response shows the designed 2 Hz and zeta 0.8. The clamp and the rate limit act at their limits. A large push held for 2 seconds and then removed returns without a long overshoot.
5. **Simulation of the whole loop.** Plant = wrist model (J, B, K, theta_0) plus servo model (second-order lag with your measured f_s, zeta_s and delay T_d). Run it at the corners of the expected ranges. Confirm it is stable wherever the two rules are met. Run one case that breaks a rule, so that you recognise what it looks like later on the hardware.

Pass. All unit tests pass and the simulation is stable at the corners where the rules hold. Keep the simulated step responses; you compare them with the real ones in Step 7.3.

If not. Simulation unstable where it should not be: check the sign of the feedback first (the wrist's reaction must act as negative feedback), then the units of T_d (milliseconds against seconds), then the discretisation, then the estimator gain.

Time. 1 day.

Step 7.2 Dry run on the hardware with the output disabled (Thu 12 Nov, morning)

Why now. It catches a wrong sign without any motion.

How. Mount the rig in configuration B. Run the full loop but do not send the servo goal; log what it would send. Push the rig by hand in each direction.

Pass. Pushing in the positive direction gives a positive torque estimate and a theta_ref that moves in the positive direction (an admittance controller yields to the push). Both axes behave the same way. Nothing saturates at rest.

If not. Wrong direction: one sign in the estimator or in the angle convention (go back to Step 2.4). Noise on the estimate at rest: look at the encoder resolution and the 8 to 10 Hz filter before changing any gain.

Time. 2 hours.

Step 7.3 First closed-loop motion on the rig (Thu 12 Nov, afternoon)

Start at the design point: K_d = 1.6 N·m/rad (natural frequency 2 Hz), rig configuration B, low rate limit, E-stop in hand, second person present.

How.

1. Hold the rig still with tau_intent = 0. The device should do nothing and stay at theta_0.
2. Push the rig gently by hand. The device should yield smoothly and return when you let go, with no chatter.
3. Inject a test torque in software: tau_intent steps of ±0.2 N·m for about 1 second (bounded, and still passed through the safety stack). Record theta_j.

4. Compare with the prediction. The reduction of the block diagram gives the steady state directly: theta_j = tau_intent / (K_d + K_h). For K_d = 1.6, K_h of about 1 and 0.2 N·m that is about 0.077 rad, roughly 4.4°. Also overlay the simulated step response from Step 7.1.

Proposed acceptance (your protocol does not fix this yet; decide it before the run): steady state within 15% of the prediction and the response shape (peak time, overshoot) within 25% of the simulation.

If not. A steady state that is off by a constant factor: k_theta is wrong (an error in the torque estimate acts like a changed K_d), so re-check Test 0. Right steady state but wrong dynamics: the servo bandwidth or T_d is not what Phases 3 and 4 measured, or the rig's J. Oscillation: reduce K_d or raise M_d and check the rule table. Chatter at rest: noise in the estimate being amplified; check the low-pass and the encoder readings. Stop immediately and cut power if the motion grows.

Time. 4 hours.

Step 7.4 Walk the allowed K_d levels, then soak (Fri 13 Nov)

How.

1. For each rig configuration (A, B, C), run Step 7.3 at each K_d level that the rule table from Step 4.5 allows (0.4, 0.8, 1.6, 3.2, 6.4 N·m/rad, subject to the rules). Move up one level at a time. Record the step response each time.
2. Optional and only on the rig: run one level that breaks a rule with a very low rate limit to see how the failure looks. Do this last, with the E-stop in hand.
3. Soak: a 30-minute run with random hand pushes. Watch the servo temperatures, the loop timing overruns and the encoder zero for drift.

Pass. Every allowed combination is stable and matches the prediction to the proposed tolerance; the soak ends with zero overruns, no zero drift and servo temperatures steady and below the datasheet limit.

If not. Overheating: the servo is holding against a spring continuously, so reduce duty or the holding offset. Drift: encoder zero or cable creep; re-tension and re-zero. Overruns during the soak that did not appear in the 60-second Test 7: a thermal or background-process effect, repeat Test 7 for 30 minutes.

Time. 1 day.

Phase 8: Safety verification before any person (Sat 14 Nov)

Gate G8: every fault in the table below produces its expected response on the rig, the limits for a person are written down and approved by your supervisor, a subject-held stop switch is built and tested, and the session checklist exists.

Why now. The rig forgives mistakes; a wrist does not. The question to answer is: if each part fails, what is the worst the device can do to a wrist, and does the system stop it? Prove that on the rig, one fault at a time.

Step 8.1 Write the limits down in numbers

How. Fix, with your supervisor (and the physiotherapist if possible), a limit for each of these and put it in the config:

- **Angle.** The device clamp is ±45°, but a person's range is smaller on some axes (typical healthy radial deviation is only about 20°; check the clinical reference you cite). Set each subject's limit inside their own measured pain-free range from Test 11, with a margin.
- **Speed.** A rate limit no faster than the fastest ramp in your patient protocol.
- **Torque.** A software ceiling on the estimated torque and on tau_intent (±0.8 N·m already), a servo torque limit if the servo has one, and the supply current limit. The servo's own capability (0.98 N·m rated, 2.94 N·m stall) is far above what a wrist should receive in a first session.
- **Hold time.** The longest the device may hold at a limit before releasing.
- **The safe state.** Define it: for example stop commanding new motion, ease the virtual spring to zero over 0.5 s, then release torque so the wrist is free. Choose "release" or "hold" deliberately and record why.

Pass. One written page of limits, signed off by the supervisor.

Time. 1 hour, plus the discussion.

Step 8.2 Inject each fault on the rig and confirm the response

How. Rig mounted, controller running at a low K_d, E-stop in hand. Inject one fault at a time and record what happens.

Fault	How to inject it	Expected response
Program freezes or crashes	Pause the process mid-motion	Servo holds its last (rate-limited) target; operator E-stop works; note the result from Step 1.4
Loop overruns repeatedly	Add a deliberate delay in the loop	Overrun counter trips and the safe state is entered
Joint encoder lost	Unplug it, then separately freeze its value	Detected within a few ticks; safe state
Encoder jumps	Spoof a jump larger than the rate limit allows	Detected as implausible; safe state
Servo bus errors	Unplug the bus briefly, then reconnect	Read failures counted; safe state after the limit
Torque estimate saturates	Push the rig hard against the device	Clamp and ceiling act; the commanded angle does not run away
Angle limit	Push the rig to the limit	Clamp holds before the rig stop; anti-windup prevents a lurch on release

Fault	How to inject it	Expected response
Sudden release	Load the rig with a weight and release it	No whip; overshoot stays inside the limits at the highest allowed K_d
Over-temperature or low voltage	Read the thresholds from the servo	Stop when the threshold is crossed
Power cut	Press the E-stop	As recorded in Step 1.4; torque removed
Operator abort	Press the software stop key	Safe state within one second

Pass. Every row gives the expected response, repeated three times, and is logged.

If not. Fix the cause and re-test the whole row and the rows near it. Do not continue past a failing row: an untested safety path counts as a missing one.

Time. 3 hours.

Step 8.3 Build the subject stop switch and write the session checklist

How.

1. Build a hand-held stop switch for the subject that cuts motor power, separate from the operator's E-stop. Test it every session.
2. Write the checklist (print it): device inspected; straps checked; config version and limits loaded; E-stop and subject stop tested; neutral zero checked; subject briefed (what will happen, how to stop, they may stop at any time); two people present with roles (operator at the PC, safety person at the E-stop).
3. Write the stop rules: stop at once if the subject reports pain, if any limit is hit unexpectedly, if the loop reports an overrun, or if anything feels wrong to the safety person.

Time. 1.5 hours.

Step 8.4 Rehearse the whole session on a mannequin or a team member's arm, power off

How. Practise donning the device, the strap sequence, the neutral alignment and doffing, then the start of a session up to the first motion, with the motors unpowered. Time it. This is also where you find a strap that is awkward or a step the checklist forgot.

Pass. Two people can do it without hesitation, in a time you can repeat.

Time. 1.5 hours.

Phase 9: Approvals and the readiness gate (request Mon 26 Oct; gate Mon 16 Nov)

Gate G9: a named person (your supervisor) confirms in writing that testing on healthy volunteers may start, every item below is ticked, and you know who your volunteers are and when they come.

Why this is its own phase. It is the one phase you cannot speed up by working harder, because it depends on other people. It starts in week 1 for that reason, and it is a gate in week 4 because a person wearing the device is the point of no return.

Step 9.1 What to ask for, and from whom (start Mon 26 Oct)

I do not know your university's rules, so confirm each of these with your supervisor and do not assume:

- Whether testing on healthy volunteers (including team members) needs ethics or departmental approval, and in what form.
- A participant information sheet and consent form (what the device does, what will be measured, the right to stop at any time, how data are stored and anonymised).
- Screening criteria (healthy adults; no current wrist, hand or forearm injury or pain; skin intact where the straps sit).
- Who must be present during sessions (supervisor, a qualified person) and what insurance or liability cover applies.

If approval is slow. Chase it weekly. If it has not arrived by Fri 13 Nov, tell your supervisor now: the fallback in the troubleshooting section is to finish on the rig and write the human protocol as ready-to-run, rather than test without approval.

Step 9.2 Plan the sessions

How.

1. Recruit two or three healthy volunteers. Test 9 needs three don/doff cycles per subject, and three people give a first idea of spread.
2. Keep each session to 60 to 90 minutes with breaks.
3. Use this order in a session, from the least to the most demanding: Test 9 (mount repeatability), Test 11 (goniometer), Tests 1 and 1b, Tests 2 and 2b, Test 3, Tests 4 and 4b (rest at least 60 s between passes), and only then closed-loop assistance.
4. Give each subject a code (S01, S02...) and use only the code in file names. Keep the key linking codes to names in a separate, protected file.
5. Measure and record before the first test: finger posture, forearm rotation, the subject's passive range of motion by goniometer (it sets their angle limit), and the hand mass and size if you need them.

Time. 2 hours of planning, plus scheduling.

Step 9.3 The readiness checklist (tick every item on Mon 16 Nov, before the first volunteer)

- ☐ Gates G0 to G8 have passed and their results are in the tracker.
- ☐ Tests 6, 7, 8, 0 and 5 are recorded; the rule table from Step 4.5 exists.
- ☐ Surrogate Tests 1 to 4b and the closed-loop checks (Phase 7) are recorded and the analysis code is frozen (tagged in git).
- ☐ Every row of the fault table (Step 8.2) has passed three times.
- ☐ The limits page is signed; the subject's own angle limit is set from their measured range.
- ☐ The subject stop switch works; the E-stop works; two named people are in the room.

- ☐ Approval, information sheet and signed consent are in hand.
- ☐ The goniometer, straps, spare cable and the printed checklist are on the bench.
- ☐ The data folder, backup and subject codes are ready.

If any item is not ticked. Do not run a session. Fix that item, or say so to your supervisor, and use the time for analysis and the report instead.

Time. 2 hours on Mon 16 Nov.

Phase 10: Healthy volunteer sessions (Mon 16 pm to Fri 20 Nov)

Gate G10: the identification tests (Tests 9, 11, 1, 1b, 2, 2b, 3, 4, 4b) are recorded on each volunteer, and closed-loop assistance has run at the allowed K_d levels without a stop event.

The principle. Do the same tests in the same order as on the rig, so a surprise on a person can be compared with the rig result. Measure first with the device passive (the controller off), and switch assistance on only after you know that person's J_h and K_h. Check each result the same day, so a failed test can be repeated at the next session instead of discovered a week later.

Step 10.1 Before every session

How.

1. Run the printed checklist from Phase 8. Test the E-stop and the subject stop switch.
2. Re-zero the neutral and check the sensors read sensibly with nobody wearing it.
3. Brief the subject: what will happen, how to stop, that they may stop at any time. Ask them to relax and let the hand feel heavy; they should neither help nor resist.
4. Fit the device with the standard finger posture and forearm rotation. Write both into the header (D-18). Photograph the strap positions.
5. Measure the subject's passive range of motion with the goniometer. Set their angle limit inside it, with a margin.

Pass. Checklist complete, limits loaded and confirmed on screen before the motors are enabled.

If not. Any item missing: do not start.

Time. 30 minutes.

Step 10.2 Session A1: repeatability and goniometer agreement (Tests 9 and 11)

How.

1. At each of three complete don/doff cycles in the same session, run Test 1 and Test 2 (and record the device's neutral and end-of-stretch angles). Between cycles, fully remove and refit the device and straps.
2. Per subject, compute the standard deviation across the three mounts for J_h, B_h, K_h and for the reference angles (the neutral and the end of stretch). Three mounts give a very uncertain standard deviation (about ±50%), so pool the within-subject variances across subjects (the square root of their mean) before computing the 95% minimum detectable change, 1.96 × sqrt(2) × the pooled standard deviation, and give its confidence interval. With three subjects the

pooled estimate has six degrees of freedom, about ±29%. Also record a reference angle that does not depend on the device (the jig stop or a digital inclinometer), because the device cannot check its own zero.

3. Test 11: measure the range with the device and, at the same time, with a manual goniometer across the range. Plot the difference against the mean (Bland-Altman) and report the bias and the 95% limits of agreement. Your tracker has no numeric limit for this yet; agree one with your supervisor before the first session (your notes tie it to an MDC of 2°).

Pass. Report the pooled MDC with its 95% confidence interval and compare it with the 2° target (ideally 1°). With nine mounts the interval is wide, so a result close to the target means "not distinguishable from it", not a clean pass or fail. The phantom result from Step 5.4 is the hardware-only floor for this number. Test 11 is reported against the limit you agreed.

If not. Check, before blaming the electronics: the strap (slip is the usual cause of mount-to-mount differences), the neutral-offset step and the finger constraint. Report the real number anyway.

Time. 75 minutes.

Step 10.3 Session A2: dynamics and spasticity checks (Tests 1b, 3, 4, 4b)

How. Same preparation. Run Test 1b once with the added masses, Test 3 (pseudo-random perturbation), Test 4 (two passes per speed, random speed order, at least 60 seconds of rest), and Test 4b (ten consecutive passes at one speed, then a rest and one more pass to see recovery). Start with the slowest speeds. Check each result against its criterion before the subject leaves.

What to expect on a healthy wrist. b_s close to zero (Test 4), and a small decay over the passes (Test 4b). A healthy subject who shows clear spasticity usually means a tense subject or an error in the passive model, so re-check Tests 1 to 3 first.

Pass. Each test meets its criterion in your test table (Block 4 tests).

If not. Use the recovery action written for that test in the Block 4 table; the main ones are: low coherence means the subject is reacting (re-brief, relax); trial spread above 20% means repeat, do not average; a disagreement above 25% between tests is a finding, so check posture, mount and transmission coupling.

Time. 75 minutes.

Step 10.4 Session B: closed-loop assistance

Why only now. You now know this person's J_h and K_h, so you can apply the rule check for them instead of hoping.

How.

1. Compute the allowed K_d levels for this subject from their measured J_h and K_h (M_d at least 3 J_h; K_d at least 0.6 K_h). Raise M_d if the J_h rule requires it, and re-derive K_d at the chosen natural frequency.
2. Start at the lowest allowed level, tau_intent = 0, and a low rate limit. The device should hold the neutral and do nothing.

3. Ask the subject to push gently. The device should yield smoothly and return.
4. Move up one allowed level at a time. At each, record the response, ask the subject for a comfort score from 0 to 10, and stop if it exceeds the level agreed in advance.
5. At each level compare the steady state with tau_intent / (K_d + K_h), using the subject's own K_h from Test 2.

Pass. Stable at every level used, no stop events, steady state within the tolerance you set in Step 7.3, and no unexpected discomfort.

If not. Any oscillation or growth: stop, cut power, and review the log before the next try. Frequent stops: strap slip or a tense subject. A tense subject raises the effective K_h and J_h, so the rule check can fail mid-session; rest and re-measure.

Time. 60 minutes.

Step 10.5 After every session

How. Doff, check the skin where the straps sat, and ask how it felt. Copy the data off the PC immediately and back it up. Run the quick analysis the same day. Fill the tracker (result, pass or fail, file name, code version). Note anything odd and photograph the final setup. Clean the straps.

Time. 45 minutes.

Suggested use of the week. Mon 16 pm: A1 on S01. Tue 17: A2 on S01, A1 on S02. Wed 18: A2 on S02, A1 and A2 on S03 if available. Thu 19: B on S01 and S02. Fri 20: B on S03, and repeat anything that failed. Sunday 15 and 22 Nov are catch-up days.

Phase 11: The intent path with replayed EMG (Tue 17 Nov, morning, on the rig)

Gate G11: the loop stays stable for every intent signal inside the bound, bad or stale intent input is rejected safely, and the end-to-end latency of the intent path is measured.

Why last, and why on the rig. The intent torque enters the sum as a feedforward term, outside the stability loop (D-05, D-12). So the loop must be proven stable and safe with the intent set to zero first (Phases 7 to 10). This phase then shows that adding intent cannot break it. Do it on the rig before a person wears it.

What this phase can and cannot show. A replayed recording shows that the pipeline and the loop behave correctly for realistic intent signals, and that stability does not depend on what the intent signal looks like. It does not show that live signals can be acquired from a patient's wrist; say that plainly in the report (D-26, D-27).

Step 11.1 Agree the interface with the ML member

How. Fix, in writing: the file format and sample rate of the replayed EMG, the timestamps, how often a new intent estimate arrives, the units and sign of tau_intent, and what the estimator outputs when it is unsure (it should output zero, not hold the last value). Replay the file at its true sample rate, as D-26 states.

Pass. A one-page interface note both of you have read.

Time. 1 hour.

Step 11.2 Check the bound and the stability with synthetic intent

How.

1. Rig configuration B, K_d = 1.6, rate limit low, E-stop in hand.
2. Apply constant intent values (0.2, 0.5 and 0.8 N·m, both signs) and confirm the steady state matches tau_intent / (K_d + K_h).
3. Apply square waves at the maximum amplitude (±0.8 N·m) at several frequencies up to about 10 Hz, including sudden sign flips. The device must stay stable and obey the angle clamp and rate limit.
4. Feed bad inputs one at a time: NaN, a value far above the bound, a value that stops updating, a burst of rapid changes. Each must be rejected or clipped, and the loop must continue safely.

Pass. Stable in every case, the bound enforced on every input, and no bad value ever reaches the servo.

If not. Anything that gets through the bound: the bound is applied after some other step; move it to the input of the sum. Oscillation at a particular frequency: that frequency is exciting the loop, which points at the measured servo bandwidth or T_d; re-check Tests 7 and 8 and the margin calculation.

Time. 2 hours.

Step 11.3 Replay recorded EMG through the real pipeline

How. Stream the recording at its true sample rate through the processing, the classifier and into tau_intent, with the loop running on the rig. Log the raw samples, the features, the estimated intent, the applied intent and the joint angle together. Measure the end-to-end latency from an EMG sample to the change in tau_intent applied, and report its distribution (median and worst case).

Add a staleness rule. If no new intent estimate arrives within a set time (choose it with the ML member), ramp tau_intent to zero rather than holding the last value. Test it by pausing the replay.

Pass. The pipeline runs for the length of the recording with no overruns, the latency is measured and recorded, and pausing the replay brings the intent to zero.

If not. Overruns when the pipeline is added: it is competing with the control loop for the processor. Move it to a separate process or core and send only the bounded intent value across.

Time. 3 hours.

Step 11.4 Write down what was shown

One paragraph in the report: what was replayed, at what rate, what the result was, and the sentence that it does not demonstrate live acquisition. This protects you in the review.

Phase 12: Results, analysis and freeze (Sat 21 to Thu 26 Nov; submit Fri 27 Nov)

Gate G12: every row of the results tracker holds a number, a pass or fail, a file name and a code version (or a written reason it was not run), the code and data are frozen and archived, and the report is written from those results.

Rule for this week: no new experiments after midday on Thu 26 Nov, and no code changes after Mon 23 Nov except bug fixes. Every number in the report must come from the tagged commit and the raw files.

Step 12.1 Complete the tracker (Sat 21 Nov)

How. For each test (0, 1, 1b, 2, 2b, 3, 4, 4b, 5, 6, 7, 8, 9, 11, and the rule check) fill: the criterion, the rig result, the human result, pass or fail, the file names, the date and the code version. A test that failed is a result: record the number, the recovery you applied and the result of the repeat. A test not run gets a one-line reason, not a blank.

Pass. No cell reads "To be measured".

Time. 4 hours.

Step 12.2 Final analysis and figures (Sat 21 to Mon 23 Nov)

How.

1. Generate every figure from the raw data with a script in the repository; do not edit images by hand. The useful set: measured against predicted step response; the Test 0 curve with hysteresis; the Test 2 stiffness loops; the Bland-Altman plot for Test 11; the Test 9 table of standard deviations and MDC; the Test 7 timing histogram; the rig results against the rig datasheet.
2. Report the number of subjects, the mean and the spread for each parameter. With two or three subjects, describe; do not claim statistical significance.
3. Show predicted against measured wherever you can. That is the chain the whole project rests on: the model predicted it, the test measured it, and here is how they differ.
4. Tag the commit and copy the raw data to an archive that is never edited.

Time. 2 to 3 days.

Step 12.3 Update the decision log and the handbook (Tue 24 Nov)

How. Add an entry for every change that came out of testing: the loop rate if it changed, the M_d policy for heavy hands, the subject angle limits, the intent staleness rule, any servo constant that replaced an assumed one. Correct the handbook where it still states an assumed value (for example the servo bandwidth and T_d) or the older FE torque description. Record what is still open: the numeric limit for Test 11, the clinical MDC, live EMG acquisition.

Time. 4 hours.

Step 12.4 Write the report and prepare the demonstration (Tue 24 to Thu 26 Nov)

How.

1. **Story to tell.** The human is in the loop, so we modelled the wrist; the model gave us rules and a list of parameters; the tests measured them on a rig and on volunteers; this is what matched and what did not; this is what we changed.

2. **Say the limits yourself,** before a reviewer does: FE torque is uncalibrated; the intent path was demonstrated with replayed recordings, not live acquisition from a wrist; healthy volunteers, not patients; small sample.
3. **Demonstration.** Rehearse it on the rig on Wed 25 Nov with the exact files and settings you will use. Record a video of a clean run as a backup in case hardware misbehaves on the day. Pack a spare cable and tools.
4. **Freeze.** Print or export the report, store the final code and data archive in two places.

Time. 3 days.

Step 12.5 Submit (Fri 27 Nov)

Keep the morning for submission and for any formatting problem. Do not plan experiments for this day.

Troubleshooting map and what to cut if time runs out

How to find a fault, in any phase

1. **What changed since the last time it worked?** Check the git difference, the config, the wiring and anything mechanical you touched. Most faults sit in the last change.
2. **Does it happen every time?** A fault that comes and goes is usually contact, timing or temperature; a fault that is constant is usually a sign, a unit or a wrong constant.
3. **Isolate by layer, from the bottom up:** mechanical, then power, then communication, then sensors, then software, then the model. Test each layer with an input you know (move the joint by hand, read the raw value).
4. **Change one thing, then re-test the failing step and the step before it.**
5. **Write down what you found,** including the wrong guesses. You will meet the same fault again.

Symptom to first checks

Symptom	Look first	Then
Servo does not answer	Servo power, then baud rate and ID	Adapter wiring and direction control, common ground, cable seating
Servo buzzes or chatters at hold	Mechanical binding or backlash	Supply voltage sag under load; noise in the torque estimate being amplified
Joint angle jumps	Magnet gap and centring	I2C wiring and pull-ups; unwrap logic
Torque estimate drifts	Cable creep and re-tension	Zero offset; temperature; strap or clamp slip
Loop overruns	Slowest call in the stage timings	USB adapter latency; grouped reads and writes; process priority; garbage collection
Rig result disagrees with its datasheet	Sign and units	Fit window; servo loading the swing; the datasheet itself

Symptom	Look first	Then
Steady state off by a constant factor	k_theta (Test 0)	Sign of the estimator; the radii ratio
Closed loop oscillates	Rule check for that K_d, M_d and the subject	T_d and servo bandwidth values; the low-pass; lower K_d or raise M_d
Large mount-to-mount spread	Strap slip	Neutral-offset step; finger posture; subject tension
Human data much noisier than the rig	Subject tension and briefing	Posture, rest, strap, time of day

Decision points in the calendar

- **Fri 30 Oct (G3).** If Test 7 fails after the recovery list, take the architecture question to your supervisor the same day and record the decision.
- **Wed 4 Nov (G4).** If Test 6 (shell rigidity) fails, stop and stiffen; this can cost days, and it moves everything after it.
- **Fri 13 Nov.** If approval for volunteers has not arrived, tell your supervisor and switch to the fallback below.
- **Mon 16 Nov (G9).** The readiness checklist decides whether anyone wears the device.
- **Thu 26 Nov, midday.** The last moment for any experiment.

If you fall behind: what to cut, in order

There is almost no slack: only the Sundays (1, 8, 15 and 22 Nov) and a few spare hours. If a gate finishes more than a day late, cut from the top of this list. Never shorten the safety phase, the readiness gate or the last three days of writing.

1. The optional rule-breaking run on the rig (Step 7.4).
2. The third volunteer.
3. Test 4b on volunteers (keep it on the rig).
4. Test 3 on volunteers (keep it on the rig).
5. The distribution of intent-path latency (keep the stability and bound checks).
6. Closed-loop assistance on volunteers (keep it on the rig).

Keep at all costs: Tests 7, 8, 0 and 6; the surrogate Tests 1 and 2 and the closed loop on the rig; the fault table; the report.

Fallback if human testing is not approved in time

Finish everything on the rig, and write the human protocol so that it is ready to run (the session plan, the checklist, the analysis scripts already tested on rig data). Say plainly in the report that human testing was not done and why. The design is still validated against a known reference, which is what D-16 asks for first; it is a smaller claim, and an honest one.

Daily routine

Start of the day. Check the backup ran, check git status is clean or understood, power up in the same order every day (supply with its current limit, then electronics, then servos), re-check the neutral zero, run the 2-minute sensor sanity log, and write the day's goal in one line.

End of the day. Power down in reverse order, back up data and code, update the tracker, note what you will do first tomorrow.